

llmXive Automated Review of TUA-Bench: A Benchmark for General-Purpose Terminal-Use Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a benchmark with specific quantitative claims that require verification against the provided data and internal consistency.

1. Task Selection and Difficulty Claims (Section 4.1 vs. Results): The authors claim in Section 4.1 that the 100 everyday tasks were selected to have the “lowest solvability” across GPT-5.5, Claude Opus 4.7, and Gemini 3.1 Pro. However, the results in Table 2 show that GPT-5.5 achieves a 60.1% success rate and Claude Opus 4.8 achieves 59.7% on the Terminus-2 agent. If the tasks were rigorously filtered for “lowest solvability” (implying near-zero or very low success rates for strong models), a ~60% success rate for frontier models suggests the selection criteria might not have been as strict as described, or the term “lowest solvability” is relative to a much larger, easier pool (which is not quantified). The claim of “lowest solvability” is strong and potentially misleading given the high absolute performance numbers reported.

2. Statistical Inconsistency in Task Distribution (Section 4.3): The text states there are 120 tasks total. It breaks them down into 100 everyday and 20 professional. In Section 4.3, the distribution is given as:

- Office & Productivity: 38.3% (~46 tasks)
- Web & Information: 18.3% (~22 tasks)
- Multimedia & Design: 13.3% (~16 tasks)

Sum of these three: 69.9% (~84 tasks). The remaining 30.1% (~36 tasks) must be “System & Software Operations” and “Scientific & Engineering”. However, the text states there are only 20 “Professional Scientific Tasks” (which presumably cover Scientific & Engineering and perhaps some System tasks). If the 20 professional tasks are the *only* ones in the Scientific & Engineering category, that is 16.7% (20/120), not the implied ~30% remainder. If “System & Software Operations” is also part of the “everyday” set, the math still doesn’t align cleanly with the stated 100/20 split unless the “System” category is large and overlaps with the “everyday” definition in a way not explicitly clarified. The percentages provided (38.3, 18.3, 13.3) do not sum to 100% with the remaining categories clearly defined, creating ambiguity in the task composition.

3. Citation and Model Version Consistency: The paper cites `openai2026introducinggpt55` and `anthropic2026opus47` in the bibliography. The text refers to “GPT-5.5” and “Claude Opus 4.8” (and 4.7). The citation `anthropic2026opus47` corresponds to Opus 4.7, but the main results table highlights Opus 4.8. The text mentions “Claude Opus 4.8” in the abstract and results, but the bibliography entry `anthropic2026opus47` is for 4.7. There is no bibliography entry for `anthropic2026opus48` (or similar) to support the 4.8 claim, although `anthropic2026claudecode` is cited. This suggests a potential mismatch between the cited source and the specific model version claimed to be evaluated, or a missing citation for the 4.8 model.

4. Cost Claims: The text claims “Claude Code + Opus 4.8 (65.8% at \$173.61/run)”. This is a very specific number. While not a “fatal” error without the raw logs, the precision implies a specific calculation method (e.g., average of 5 trials, specific token counts). The paper does

not detail the cost calculation methodology (e.g., whether it includes the cost of the agent framework overhead or just the LLM API calls), which is critical for the “Cost-performance trade-off” claim.

The paper is generally sound in its structure, but the specific numerical claims regarding task difficulty selection and the internal consistency of the task distribution percentages need clarification to ensure the “lowest solvability” claim is not an overstatement and the task counts match the reported percentages.

Action items.

- **[science]** Section 4.1 claims tasks were selected for ‘lowest solvability’ across specific models, yet results show ~60% success for frontier models. Clarify if ‘lowest’ is relative to a larger pool or if the selection criteria description is inaccurate regarding the resulting difficulty.
- **[writing]** Task distribution percentages in Section 4.3 (Office 38.3%, Web 18.3%, Multimedia 13.3%) sum to 69.9%, leaving 30.1% for two categories. However, the text states only 20 professional tasks exist (16.7%). The math for the remaining categories is inconsistent with the stated 100/20 split.
- **[writing]** The bibliography cites ‘anthropic2026opus47’ but the results table and text highlight ‘Claude Opus 4.8’ as the top performer. Add a specific citation for the Opus 4.8 model to support the claim that it was evaluated and achieved 65.8%.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 provides a clear conceptual overview of the TUA-Bench pipeline, but the embedded ‘Cost Efficiency’ and ‘Success Rate’ charts suffer from illegible text and missing axis labels, reducing their immediate interpretability.

Figure 2

Figure 2 is a clear and well-constructed dual-axis line chart that effectively visualizes the relationship between task-execution time budget, success rate, and timed-out trials. The axes are clearly labeled with units, the legend distinguishes the two data series, and specific data points are annotated for precision. The visual presentation aligns perfectly with the provided caption.

Figure 3

Figure 3 effectively visualizes success rate vs. cost with clear legends and labels, but the claimed Pareto frontier line is inaccurately drawn, the x-axis scale is implicitly logarithmic without notation, and the dual encoding of agent/model attributes lacks explicit explanation in the caption.

Figure 4

Figure 4 is visually clear with well-labeled axes and a complete legend. However, the caption contains a broken sentence referencing a missing figure, and the model version numbers in the legend appear inconsistent with the rest of the paper.

Figure 5

The heatmap effectively visualizes success rates but suffers from severe x-axis label clutter due to rotation and overlap. Additionally, the visual grouping elements (separators and task counts) described in the caption are missing from the rendered image.

Figure 6

The heatmap effectively visualizes the full task-level results across categories, but the extreme density of the x and y-axis labels renders them illegible, and the colorbar legend is missing from the image.

Figure 7

Figure 7 effectively visualizes the effect of reasoning effort on task performance across different models and categories. However, the text labels for both the x-axis (effort settings) and y-axis (task names) are rendered at an illegible size, hindering the ability to read specific data points or task details.

Action items.

- **[writing]** Figure 1: The ‘Cost Efficiency’ chart (bottom right) contains illegible text labels for data points (e.g., model names like ‘Terminus-2’, ‘Codex’, ‘Claude Code’) and axis values due to low resolution and small font size.
- **[writing]** Figure 1: The ‘Success Rate by Task Category’ bar chart (middle right) lacks a Y-axis label and scale, making the percentage values (e.g., 61%, 52%) the only reference for magnitude.
- **[science]** Figure 3: The caption claims the connected line marks the Pareto frontier, but the dotted line connects points that are not Pareto-optimal (e.g., the ‘Terminus-2 MiniMax-M3’ point at ~\$12 is connected to ‘Terminus-2 GLM-5.1’ at ~\$23, yet there are other points with higher success rates at similar or lower costs, such as the orange square at ~\$30 with ~41% success). A true Pareto frontier should only connect points where no other point has both higher success rate and lower cost.

- **[writing]** Figure 3: The x-axis label ‘Cost per run (USD)’ is clear, but the axis scale is logarithmic without explicit indication (e.g., no log-scale notation or tick labels like 10^1 , 10^2), which may mislead readers about the spacing of cost values.
- **[writing]** Figure 3: The legend defines ‘Agent’ shapes and ‘Model’ colors, but some points (e.g., the red triangle labeled ‘Claude Code’) use a shape (triangle) that corresponds to ‘Claude Code’ in the Agent legend, yet the color (red) corresponds to ‘Claude Opus 4.8’ in the Model legend — this dual encoding is not explicitly explained in the caption, potentially causing confusion about which attribute (agent or model) is being emphasized for each point.
- **[writing]** Figure 4 caption contains a broken cross-reference: ‘A more detailed, task-level breakdown of success rates is provided in .’ The target figure name or number is missing.
- **[writing]** Figure 4 legend labels include version numbers (e.g., ‘GPT-5.5’, ‘Claude Opus 4.8’) that do not match the model names in the paper title or other figure captions (e.g., ‘GPT-4’, ‘Claude 3.5’), creating potential confusion about which models are evaluated.
- **[writing]** Figure 5: The x-axis labels (task names) are rotated 45 degrees and overlap significantly, making them illegible and cluttered.
- **[writing]** Figure 5: The caption states ‘grouped by category and subcategory’ and mentions ‘n’ (number of tasks) below groups, but the rendered image lacks the vertical separator lines and the ‘n’ counts shown in the related Figure 4.
- **[writing]** Figure 6: The x-axis labels (agent + model + effort) are extremely dense, rotated, and illegible, making it impossible to distinguish specific configurations without zooming in significantly.
- **[writing]** Figure 6: The y-axis task labels are similarly dense and rotated, causing overlap and rendering many task names unreadable.
- **[science]** Figure 6: The colorbar legend is missing from the rendered image; while the caption defines the red-to-green scale, the visual key is absent from the figure itself.
- **[writing]** Figure 7: The x-axis labels for the reasoning effort settings (none, low, medium, high, xhigh, max) are rotated 90 degrees and rendered at a font size that is illegible in the provided image, making it impossible to distinguish the specific effort level for each column.
- **[writing]** Figure 7: The task identifiers and names on the y-axis are extremely small and densely packed, rendering them illegible without significant zooming, which contradicts the caption’s instruction to ‘zoom in’ to view details.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript generally avoids excessive jargon, but several terms and acronyms are used without definition, potentially excluding non-specialist readers.

First, in **Table 1** (line 108), the column header **MM** is used to denote “Multimodal” requirements. This abbreviation is not defined in the table caption or the surrounding text. While common in computer vision circles, it is opaque to a general audience. The authors should spell out “Multimodal” or define “MM” at first use.

Second, the term **Scaffold** appears frequently in the results analysis (e.g., Section 5.2, lines 235, 242) to refer to the underlying agent framework or orchestration layer (e.g., Terminus-2, OpenHands). In this context, “Scaffold” is used as a specific technical noun without definition. Replacing this with “agent framework,” “orchestration layer,” or “execution environment” would improve clarity for readers unfamiliar with this specific terminology.

Third, the metric **All-5** is introduced in Section 5.1 (line 188) and Table 2. While the table header implies a count, the text does not explicitly define it as “consistent success in all 5 trials” until the caption or later discussion. Defining this metric explicitly at its first mention in the text would prevent ambiguity.

Fourth, the Figure 1 caption (line 42) uses the term **rollout** (“the resulting rollout is automatically verified”). This is standard reinforcement learning jargon for a sequence of actions and states. For a general-purpose benchmark paper, “execution trace” or “sequence of actions” is more accessible.

Finally, in the task descriptions (e.g., Task 025, line 338), specific software internal terms like **DataPilot** (LibreOffice Calc’s pivot table feature) are used. While necessary for the task specification, the paper should ensure that such terms are either defined in a glossary or replaced with generic descriptions (e.g., “pivot table feature”) in the main text to maintain accessibility.

Action items.

- **[writing]** The manuscript generally avoids excessive jargon, but several terms and acronyms are used without definition, potentially excluding non-specialist readers. First, in Table 1 (line 108), the column header **MM** is used to denote “Multimodal” requirements. This abbreviation is not defined in the table caption or the surrounding text. While common in computer vision circles, it is opaque to a general audience. The authors should spell out “Multimodal” or define “MM” at first use. Second, the term **Scaf**

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript presents a benchmark for terminal-use agents, but there are significant logical inconsistencies between the stated task counts, distribution statistics, and the reported

experimental results.

First, in Section 3.1 and 3.2, the authors state that 100 everyday tasks and 20 professional tasks were selected, summing to 120. However, in Section 3.3 (“Task Statistics and Features”), the distribution is given as: Office & Productivity (38.3%), Web & Information (18.3%), and Multimedia & Design (13.3%). The sum of these percentages is 70%. The text fails to provide percentages for “System & Software Operations” and “Scientific & Engineering,” which are listed as families. Without these values, the claim that these five families constitute the 120 tasks is logically incomplete. Furthermore, 38.3% of 120 is 45.96 tasks, which is not an integer, suggesting the percentages are either rounded inconsistently or derived from a different total than 120.

Second, and more critically, there is a direct contradiction between the data in Table 1 (Section 4.2) and the textual analysis immediately following it. Table 1(a) shows:

- Terminus-2 + GPT-5.5: Success Rate 60.1%, All-5: 31.7%
- Terminus-2 + Claude Opus 4.8: Success Rate 59.7%, All-5: 42.5%

The text states: “GPT-5.5 (60.1%) and Claude Opus 4.8 (59.7%) are close on Terminus-2, but reliability differs (All-5: 42.5% vs 31.7%).” The phrasing “42.5% vs 31.7%” grammatically links the first value (42.5%) to the first model mentioned (GPT-5.5) and the second value (31.7%) to the second model (Opus 4.8). This is the exact opposite of the data in the table, where Opus 4.8 has the higher All-5 score (42.5%) and GPT-5.5 has the lower (31.7%). This error undermines the conclusion that Opus 4.8 shows “higher reliability” if the reader relies on the text’s explicit mapping of numbers to models. The authors must correct the text to accurately reflect the table’s data or vice versa.

Finally, in the “Ablation Studies” section, the text claims “Success rises from 36.5% (none) to 60.1% (xhigh).” While the 60.1% figure matches the GPT-5.5 result in Table 1(a), the 36.5% figure is not explicitly defined in the table or the surrounding text as a baseline for a specific model/agent configuration, making the causal claim of “thinking-effort scaling” difficult to verify without the missing figure data or a clearer textual reference to the specific agent/model pair used for this ablation.

Action items.

- **[writing]** Section 3.3 lists five task families but only provides percentages for three (summing to 70%). The missing percentages for ‘System & Software Operations’ and ‘Scientific & Engineering’ break the logical link to the claimed total of 120 tasks.
- **[science]** Section 4.2 text claims GPT-5.5 has All-5 of 42.5% and Opus 4.8 has 31.7%, but Table 1(a) shows the reverse (Opus 4.8: 42.5%, GPT-5.5: 31.7%). The text contradicts the table data it cites.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims regarding the scope and rigor of TUA-Bench that appear to exceed the evidence provided in the manuscript.

First, the assertion that the benchmark includes tasks “co-designed with PhD-level domain experts” (Abstract, Section 1, Section 3.2.2) is a significant claim regarding the validity and difficulty of the scientific tasks. However, the manuscript provides no details on the expert curation process. It does not state how many experts were involved, their specific fields of expertise, the duration of the collaboration, or the specific criteria used to validate the tasks. Without a description of the expert-in-the-loop methodology, this claim functions as an appeal to authority rather than a demonstrated scientific contribution. The authors must either provide the methodological details of this co-design process or rephrase the claim to reflect that tasks were “inspired by” or “adapted for” these domains.

Second, the paper claims to convert tasks from OSWorld to the CLI (Section 3.2.1). This implies a direct translation of GUI-based workflows to terminal commands. However, many GUI tasks rely on visual grounding (e.g., “click the red button,” “drag the file to the trash”) that has no direct CLI equivalent. The paper over-reaches by presenting these as “real-world” terminal tasks without explaining how the semantic gap was bridged. If the tasks were fundamentally altered to fit the CLI (e.g., replacing a drag-and-drop with a `mv` command), the resulting task may no longer reflect the original “real-world” complexity or intent. The authors need to clarify the extent of this conversion and whether the tasks remain faithful to the original real-world scenarios or are merely synthetic CLI exercises.

Finally, the title and abstract describe TUA-Bench as a benchmark for “General-Purpose” terminal agents. While the benchmark covers five families, the distribution is heavily weighted toward Office & Productivity (38.3%) and System & Software Operations, with only 20 tasks (16.7%) in the Scientific & Engineering category. The term “general-purpose” implies a broad capability across diverse computing domains, but the data suggests a focus on office automation and system administration. The claim of “general-purpose” is an over-extrapolation of the current task distribution. The authors should either temper this claim (e.g., “General-Purpose Office and System Agents”) or provide a stronger justification for why the current distribution represents “general-purpose” computing.

Action items.

- **[writing]** The claim that tasks are ‘co-designed with PhD-level domain experts’ (Abstract, Sec 1) is over-claimed given the lack of methodological detail. The paper does not specify the number of experts, their specific qualifications, the iterative design process, or how their input was validated. Without this, the ‘expert’ label is a marketing term rather than a scientific claim. Add a subsection detailing the expert curation protocol.
- **[science]** The statement that tasks were ‘converted to CLI’ from OSWorld (Sec 3.2.1) implies a direct mapping that may not exist. Many OSWorld tasks rely heavily on visual GUI cues (e.g., specific button locations, drag-and-drop) that do not have direct CLI equivalents. The paper over-reaches by suggesting these are ‘real-world’ terminal tasks without explaining

how the semantic gap between GUI intent and CLI execution was bridged or if the tasks were fundamentally altered.

- **[writing]** The benchmark claims to evaluate ‘general-purpose’ agents (Title, Abstract), yet the task distribution is heavily skewed toward Office/Productivity (38.3%) and System Ops, with only 20/120 tasks in Scientific/Engineering domains. The term ‘general-purpose’ suggests broad coverage across all computing domains, but the data does not support a claim of generalization beyond office automation and basic system administration. Temper the ‘general-purpose’ claim or expand the scientific task set.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper introduces TUA-Bench, a benchmark for terminal-use agents, which inherently carries safety and dual-use risks due to the nature of terminal interactions (system administration, file manipulation, and web automation). While the authors mention using isolated Linux containers (Harbor framework) in Section 3.1, the safety review requires more explicit mitigation strategies for specific high-risk tasks.

First, regarding system administration tasks, Task 107 (106-create-charles-ssh-user) explicitly instructs the agent to create a new system user with a password and home directory. While likely intended for a sandbox, the manuscript lacks a clear statement confirming that these operations are strictly confined to ephemeral containers and that no persistent changes are made to the host infrastructure. Without this explicit confirmation, there is a risk that the benchmark could be misused to test privilege escalation or persistence mechanisms on real systems. The authors should add a sentence in Section 3.1 or the Limitations section clarifying the isolation guarantees and the disposal of container states post-execution.

Second, the benchmark includes tasks interacting with live commercial websites (e.g., Task 027 adding items to a Steam cart, Task 037 searching for cars). Automated agents interacting with live sites pose risks of violating Terms of Service, triggering bot detection, or inadvertently causing financial transactions if the agent logic fails. The paper should describe the safeguards in place, such as rate limiting, the use of test accounts with zero balance, or the restriction of agents to “read-only” or “cart-add-only” modes that do not proceed to checkout.

Finally, tasks involving sensitive data handling (e.g., Task 044 navigating to saved passwords, Task 099 extracting images from email attachments) require assurance that the dataset does not contain real PII. The authors should explicitly state that all input artifacts (emails, password stores) are synthetic or fully anonymized to prevent privacy violations during the benchmarking process. Given the “English-only” limitation mentioned in Section 6, there is also a potential bias in the safety evaluation, as agents might behave differently on non-English interfaces, but this is a secondary concern compared to the direct system and privacy risks.

Action items.

- **[writing]** Task 107 (073-force-quit-frozen-doc) and Task 107 (106-create-charles-ssh-user) involve creating system users and modifying OS configurations. The paper must explicitly state that these tasks are executed in isolated, disposable containers (Harbor framework) and that no persistent system changes or real user accounts are created on host machines to prevent privilege escalation risks.
- **[writing]** Several tasks (e.g., 027-add-dota-dlc-cart, 037-electric-cars-under-50k) involve interacting with live commercial websites (Steam, Cars.com). The manuscript should clarify the protocol for rate limiting, bot detection avoidance, and ensuring that automated agents do not violate Terms of Service or incur unintended financial charges during benchmark execution.
- **[writing]** Task 044 (044-etsy-saved-passwords) and Task 099 (098-extract-email-doc-image) involve accessing sensitive user data (passwords, emails). The authors must confirm that all test data used in the benchmark is synthetic or anonymized, and that the evaluation protocol explicitly forbids agents from exfiltrating or logging real credentials.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the claims of TUA-Bench is generally robust in terms of scale (120 tasks) and execution-based evaluation, but there are gaps in statistical rigor and potential selection bias that require clarification.

Statistical Rigor and Significance: The main results in Section 4.2 (Table 1) present performance metrics with standard deviations derived from 5 independent trials (e.g., Claude Code + Opus 4.8: 65.8 ± 0.7 ; Codex + GPT-5.5: 64.7 ± 0.7). While the inclusion of error bars is positive, the paper does not report the results of statistical significance tests (e.g., paired t-tests, Wilcoxon signed-rank tests, or bootstrap confidence intervals) to determine if the observed differences between top-performing agents are statistically significant or merely artifacts of random variance in the small sample size ($N=5$). Given the small margins (e.g., 1.1% difference), this is a critical omission for a benchmark paper claiming to rank frontier models.

Selection Bias in Task Curation: In Section 3.2.1, the authors describe curating the “Everyday Digital Tasks” by selecting the 100 tasks with the “lowest solvability” across GPT-5.5, Claude Opus 4.7, and Gemini 3.1 Pro. This “hardness-based” selection introduces a significant risk of selection bias. By explicitly filtering for tasks where these specific models fail, the benchmark may inadvertently over-represent failure modes specific to the architectures or training data of those three models. This could lead to an overestimation of the difficulty for other agent frameworks or models with different inductive biases, potentially skewing the comparative results.

The authors should discuss this limitation or provide an analysis of task difficulty distribution across a broader set of models to demonstrate the benchmark’s generalizability.

Validation of Professional Tasks: The “Professional Scientific Tasks” (Section 3.2.2) are a key differentiator, yet the evidence for their quality is qualitative. The paper states they were “co-designed with PhD-level experts” and refined from 25 to 20 tasks. However, it lacks quantitative metrics on the validation process, such as inter-rater reliability among experts, the specific criteria used to reject the initial 5 tasks, or a pilot study showing that human experts can solve these tasks with high success rates. Without this, the claim that these tasks are “reliable” and “meaningful” is not fully supported by the provided evidence.

Replication and Variance: The use of 5 trials is a reasonable baseline, but for tasks involving stochastic elements (e.g., web navigation, image generation), the variance might be higher than reported. The paper should explicitly state whether the 5 trials were run with different random seeds and if the environment was reset identically each time. The ablation studies (Section 4.3) show clear trends (e.g., time budget scaling), which strengthens the internal validity, but the main comparative claims remain slightly under-supported by statistical testing.

Action items.

- **[science]** Clarify the statistical significance of the reported performance gaps. Table 1 reports means with standard deviations (e.g., 65.8 ± 0.7 vs 64.7 ± 0.7), but no hypothesis tests (e.g., paired t-tests or bootstrap confidence intervals) are provided to confirm these differences are not due to random variance across the 5 trials.
- **[science]** Address potential selection bias in the ‘Everyday Digital Tasks’ subset. The text states 100 tasks were retained based on ‘lowest solvability’ across three specific models. This curation method risks overfitting the benchmark to the failure modes of those specific models, potentially inflating the difficulty for other architectures or under-representing tasks where those models happen to succeed.
- **[science]** Provide a detailed breakdown of the ‘Professional Scientific Tasks’ validation. While the paper mentions co-design with PhD experts, it lacks quantitative inter-rater reliability metrics or a description of the validation protocol used to ensure the 20 tasks are solvable and unambiguous, which is critical for the reliability of the benchmark’s ‘depth’ claim.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in TUA-Bench is generally sound in its execution of multiple trials (n=5) and reporting of mean success rates with standard deviations. However, the manuscript lacks rigorous statistical inference to support its comparative claims.

In Section 4.1 (Main Results), Table 1 presents performance metrics with standard deviations

(e.g., Claude Code + Opus 4.8 at 65.8 ± 0.7 vs. Codex + GPT-5.5 at 64.7 ± 0.7). While the point estimates suggest a difference, the small sample size (5 trials) makes the standard error relatively large. The authors claim the former is the “highest score” but do not report p-values, confidence intervals, or the results of a statistical test (e.g., a paired t-test or bootstrap test) to confirm that this difference is statistically significant rather than a result of random variance. Without this, the ranking of the top agents is not statistically defensible.

Similarly, the ablation studies in Section 4.2 (e.g., Thinking-effort scaling) report specific percentage point gains (e.g., +2.3 points from ‘high’ to ‘xhigh’) but omit error bars or significance testing. Given the stochastic nature of LLM agents, it is crucial to determine if these gains are robust. The claim of “diminishing returns” relies on the assumption that the marginal gain is real; without statistical validation, this could be an artifact of the specific random seeds used.

Finally, the task curation methodology in Section 3.2.1 describes a selection bias where tasks were filtered to retain those with the “lowest solvability.” While this creates a challenging benchmark, the statistical implications of this selection process on the generalizability of the results are not discussed. The authors should clarify if the resulting difficulty distribution is representative or if the benchmark is skewed toward a specific failure mode, which would affect the external validity of the performance metrics.

Action items.

- **[science]** Section 4.1 and Table 1 report success rates with standard deviations (e.g., 65.8 ± 0.7) derived from 5 trials. The manuscript must explicitly state the statistical test used to determine if the difference between the top two agents (65.8% vs 64.7%) is significant, or provide confidence intervals. Without this, the claim of ‘highest score’ is not statistically supported given the small sample size ($n=5$).
- **[science]** The ablation study on ‘Thinking-effort scaling’ (Section 4.2) reports monotonic gains (36.5% to 60.1%) but lacks error bars or significance testing. Given the high variance often seen in LLM benchmarks, the authors must verify if the gain from ‘high’ to ‘xhigh’ is statistically distinguishable from noise before claiming ‘diminishing returns’.
- **[science]** The task selection process for the ‘Everyday Digital Tasks’ (Section 3.2.1) states tasks were retained based on ‘lowest solvability’ across three models. The authors must clarify if this selection introduced bias (e.g., selecting only tasks where models failed consistently) and if the resulting distribution of difficulty is representative of the broader domain, or if it artificially inflates the benchmark’s difficulty.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a generally high standard of academic writing, with a clear logical

flow from the introduction of the benchmark to the experimental results. The abstract effectively summarizes the scope and key findings. However, several areas require refinement to enhance precision and readability.

In Section 3.1 (“Task Curation”), the description of the curation process lacks specific detail regarding the verification step. The phrase “Rigorous human verification removes input-gold mismatches” is too generic. To improve clarity, the authors should briefly specify the protocol used (e.g., “Two independent annotators verified...”). This addition would strengthen the credibility of the benchmark construction without adding significant length.

In Section 4.2 (“Main results”), the comparison between models contains a minor ambiguity. The sentence “GPT-5.5 (60.1%) and Claude Opus 4.8 (59.7%) are close on Terminus-2, but reliability differs” leaves the reader to infer which metric defines “reliability.” While the subsequent mention of “All-5” clarifies this, integrating the metric name directly into the comparative sentence (e.g., “...but reliability differs in the All-5 metric”) would improve immediate comprehension.

A more significant issue affects the Appendix, specifically the “Full Task List” section. Several task descriptions (e.g., Task 082, 098, 106) contain unresolved placeholders such as “{path}” or generic references like “the input video.” Since this section serves as a concrete example of the benchmark’s content, these placeholders disrupt the reading flow and reduce the utility of the appendix. The authors should replace these with specific, representative file paths or a consistent, clearly defined notation (e.g., `<input_file>`) to ensure the text is self-contained and professional.

Overall, the paper is well-structured, but addressing these specific clarity and consistency issues will significantly improve the final presentation.

Action items.

- **[writing]** In Section 3.1 (Task Curation), the sentence ‘Rigorous human verification removes input-gold mismatches’ is vague. Specify the verification protocol (e.g., ‘Two independent annotators verified...’) to improve clarity and reproducibility.
- **[writing]** Section 4.2 (Main results) contains inconsistent phrasing: ‘GPT-5.5 (60.1%) and Claude Opus 4.8 (59.7%) are close on Terminus-2, but reliability differs’. Clarify that the reliability difference refers to the ‘All-5’ metric explicitly in the same sentence to avoid ambiguity.
- **[writing]** The Appendix task descriptions (e.g., Task 082, 098, 106) contain placeholder text like ‘{path}’ instead of specific file paths. Replace these with concrete examples or a consistent notation (e.g., ‘<input_path>’) to ensure the text is self-contained and readable.